7 (a) Write the number 0.000 086 23 in standard form.

Move the decimal 5 places to get 8.623.

8.623×10^{-5}

1 mark
Number between 1 and 10; exponent −5

(1)

(b) Work out $\dfrac{3.2 \times 10^{3} + 5.1 \times 10^{-2}}{4.3 \times 10^{-4}}$

Give your answer in standard form, correct to 3 significant figures.

$\dfrac{3200 + 0.051}{0.00043} = 7\,441\,979.0697\ldots$

1 mark
Evaluate the whole fraction

$= 7.441979\ldots \times 10^{6} \approx 7.44 \times 10^{6}$

7.44×10^{6}

1 mark
Standard form to 3 significant figures

(2)

(Total for Question 7 is 3 marks)